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Agentic AI (IA462) , Semester VIII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_1_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 1

Simple Reflex (Reactive) Agent — The Vacuum-Cleaner World

Aim

To implement a reactive agent that maps percepts directly to actions through condition–action rules, and to measure its behaviour with a performance score in a simulated environment.

Theory

An agent is anything that perceives its environment through sensors and acts upon it through actuators. The simplest architecture is the simple reflex agent: a fixed set of condition–action rules maps the current percept straight to an action, with no internal state, no model of the world and no look-ahead. Russell and Norvig's vacuum-cleaner world is the canonical testbed: two locations that may be dirty, an agent that can move and suck, and a performance measure that rewards cleaning and penalises movement.

The experiment also establishes the agent–environment separation that every later experiment reuses: the environment owns the true state and exposes only percept() and execute(action); the agent sees nothing but the percept. Note the limitation visible in the sample output — once both squares are clean the reflex agent oscillates forever, bleeding performance, because it cannot remember that the other square was already clean. Fixing this requires internal state (Experiment 2).

Table 1. PEAS description of the task environment.

PEAS component	Specification for this experiment
Performance measure	+1 for each clean square at each time step; −1 for every movement (energy cost)
Environment	Two squares A and B, each Clean or Dirty; deterministic, fully observable locally
Actuators	Suck, Left, Right, NoOp
Sensors	Location sensor and dirt sensor, giving the percept (location, status)

Algorithm

1. Build an environment class holding the dirt status of both locations, the agent's location and a performance score.
2. Implement percept() returning (location, status) and execute(action) applying suck/left/right with rewards +10 and −1.
3. Implement the agent as a pure function: if dirty then suck, else move to the other square.
4. Run the perception–action loop for a fixed number of steps and report the final state and score.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the simple reflex agent below, including the initial state, the percept–action trace of every step, and the final state with the performance score.

Observations

Comment on why the agent keeps moving after both squares are clean, on how the performance score changes with the number of steps, and on what the result tells you about the need for internal state.

Exercises

1. Add a model-based variant that remembers the status of both squares and executes a no-op once everything is known to be clean; compare scores over 20 steps.
2. Extend the world to a 1×N corridor of N squares with random initial dirt and adapt the rules.
3. Make dirt reappear with probability 0.1 per step and discuss whether the reflex agent is now rational.